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Teaching self-questioning using systematic prompt fading: Effects on fourth graders' reading comprehension

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ABSTRACT

Teaching students to generate questions to monitor their reading comprehension is an evidence-based practice demonstrated to be effective across grade levels for students with and without disabilities. As self-questioning is a complex skill, many students can benefit from additional instructional supports, especially when they are first learning how to self-question. One such support is systematic prompt fading, an evidence-based practice in which instructional prompts are gradually faded until the student can perform the skill independently. This study used systematic prompt fading to teach fourth graders a self-questioning strategy to improve their reading comprehension. A multiple probe across participants design demonstrated improved comprehension quiz scores for fourth graders who engaged in self-questioning while reading expository passages. In addition, all participants demonstrated at least some generalization to higher-level content area textbooks.

KEYWORDS

At-risk; reading comprehension; self-questioning; teaching self-questioning using systematic prompt fading

Reading comprehension is an important skill for success in school and throughout life. As students advance through each grade level in school, they are expected to read increasingly more content that becomes more complex and difficult for struggling readers (e.g., Gajria, Jitendra, Sood, & Sacks, 2007; Solis, Ciullo, Vaughn, Pyle, Hassaram, & Leroux, 2012). As the demands of reading comprehension increase, the achievement gap between proficient and nonproficient readers continues to expand. Content areas such as social studies and science require a significant amount of independent reading and, consequently, students who struggle with reading comprehension will likely struggle in content areas as well. When implementing instruction to support reading comprehension for students who are at risk for reading failure, it is critical for teachers to use evidence-based practices.

One evidence-based practice with decades of empirical support is teaching students to use self-questioning to monitor their reading comprehension (Joseph, Alber-Morgan, Cullen, & Rouse, 2015; Rosenshine, Meister, & Chapman, 1996; Wong, 1985). Self-questioning is a reading comprehension strategy that requires students to stop periodically while reading (e.g., at the end of each paragraph) and to ask and answer questions about the content they read. For example, Berkeley, Marshak, Mastropieri, and Scruggs (2011) taught seventh graders in a general education classroom to generate their own questions by looking at headings in a reading passage, making the headings into questions, reading the passage, and stopping periodically to ask and answer their self-generated questions. For example, if students were given the heading "The Election of 1976", they would generate questions such as "Who won the election?" or "What was important about the election?" Results demonstrated that students who received self-questioning instruction outperformed students in the control group on reading comprehension of social studies content.

Two early literature reviews on self-questioning (Rosenshine et al., 1996; Wong, 1985) provided evidence that self-questioning interventions are generally effective for improving reading comprehension of students in third grade through college. Both of these literature reviews included only studies that used control group comparison designs. Most of the studies targeted students without disabilities attending general education classrooms. More recently, Joseph et al. (2015) reviewed the self-questioning research from 1990 to 2012. Results of Joseph et al. support the findings of the previous two literature reviews. In addition, Joseph et al.'s review identified interventions from several studies specifically targeting students with disabilities or students at risk for reading failure. Effective interventions included variations of self-questioning intervention packages such mnemonic learning strategies, graphic organizers, embedded story structure, and self-recording. For example, the RAP strategy (Read a paragraph, Ask yourself the main idea and two details, Paraphrase the main idea and details in your own words) was effective for improving reading comprehension of at-risk sixth graders (Hagaman & Reid, 2008) and at-risk third graders (Hagaman, Casey & Reid, 2010). Other mnemonic self-questioning strategies demonstrated to be effective included QAR (Question Answer

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Relationships; Vaughn et al., 2011), TWA (Think before you read, While you read, After you read; Johnson et al., 2011), RAAC (Reread-Adapt and Answer-Comprehend; Therrien, Wickstrom, & Jones 2006), and ART (Ask, Read with Alertness, Tell; McCallum et al., 2011).

In about one-third of the studies reviewed by Joseph et al. (2015), students were not taught to generate their own questions; rather, the teacher provided the questions (e.g., Crabtree et al., 2010; Taylor et al., 2002; Therrien et al., 2006). For example, Taylor et al. (2002) and Crabtree et al. (2010) provided a list of generic questions related to narrative story elements (e.g., Who are the main characters? What is the setting? What is the conflict?). The students were taught to stop at three marked stopping points in the story to ask and answer the questions and record their responses on a self-monitoring form (i.e., Crabtree et al.) or an audio recorder (i.e., Taylor et al.). Although students substantially improved their reading comprehension using self-questioning with teacher-provided questions, they were not taught to generate their own questions. Question generation is arguably an advanced skill with which many at-risk learners may struggle a great deal. Therefore, providing a systematic way for students to transition from using teacher-prepared questions to generating their own questions can increase their independence and generalization to new reading materials. One way to accomplish this goal is to systematically fade instructional prompts until students can perform the skill independently.

Although systematic prompt fading was demonstrated to be effective for teaching sight words (e.g., Richardson et al., 2017; Rivera, Koorland, & Fueyo, 2002), the authors only found one published research study examining prompt fading to teach self-questioning for reading comprehension. Rouse, Alber-Morgan, Cullen, and Sawyer (2014) used systematic fading of embedded questions to teach fifth graders to self-generate questions. Specifically, students were first provided with a passage with an embedded question after each paragraph accompanied by a blank line to write the answer. The embedded questions were gradually faded (i.e., one at a time in each subsequent reading passage) and replaced with blank lines for the students to write both a self-generated question and an answer. Then the blank lines were gradually faded and replaced with small pictures of stop signs to prompt the students to orally self-question and answer. In the maintenance phase, the stop sign prompts were removed from the passages and the students were expected to draw their own stop signs at the end of each paragraph, read the passage, and self-question independently. Results of this study demonstrated significant improvements on reading comprehension as well as maintenance for each participant.

The current study, a systematic replication of Rouse et al. (2014), was designed to extend previous research by examining the effects of teaching self-questioning with systematic prompt fading to at-risk fourth graders. In contrast to Rouse et al., the embedded questions condition was not implemented in this study. After baseline, the experimenters immediately taught the students to self-generate questions (through modeling, guided practice, and feedback) and write their questions and answers on blank lines embedded after each paragraph in a reading passage. Similar to Rouse et al., we gradually faded the blank lines to stop sign prompts.

A limitation of Rouse et al. was that generalization probes were not administered during baseline; they were only administered during the maintenance phase. This precluded Rouse et al. from drawing conclusions about the effectiveness of the intervention on generalization. In the current study, the experimenters collected generalization probes during baseline and intervention in order to experimentally determine whether a functional relation was demonstrated. The experimenters (i.e., a doctoral student at the time of the study and a professor) designed this study to answer the following research questions. What are the effects of a self-questioning prompt fading package on the reading comprehension of fourth graders? What are the effects of a self-questioning prompt fading package on generalization to higher-grade-level reading materials?

Table 1. Student information.

Participants	Gender	Ethnicity	AIMSweb oral reading fluency	AIMSweb maze
Nellie	F	Hispanic	128	14
Edith	F	White	130	18
Ansley	F	Multiracial	120	7

Method

Participants. The participants were three fourth grade girls who were identified as at-risk readers by their language arts teacher. Nellie was a Hispanic girl who scored 128 words per minute on AIMSweb oral reading fluency and 14 on AIMSweb maze. Edith was a White girl who scored 130 on oral reading fluency and 18 on maze. Ansley was a multiracial girl who scored 120 on oral reading fluency and 7 on maze. Based on their reading fluency and maze scores at the beginning of the study, all three participants demonstrated on–grade level oral reading fluency and below–grade level reading comprehension. See Table 1 for student information and AIMSweb oral reading fluency and maze scores.

Setting. This study was conducted in a suburban elementary school in a Midwestern state. Data were collected in the language arts/reading classroom on four to five days per week during 15- to 20-minute one-on-one sessions. During baseline and intervention sessions, there were usually 25 students present in the classroom as well as the classroom teacher, the experimenter (i.e., first author) and, periodically, an additional observer. The experimenter, who was a doctoral student with 10 years of teaching experience at the time of the study, implemented all baseline and intervention sessions. The experimenter conducted sessions at a table in a quiet area of the classroom following a period of teacher-led whole-class instruction. During intervention sessions, the experimenter worked with each participant individually while the teacher led small-group instruction and/or monitored independent work.

Materials

Expository reading passages. The experimenters obtained expository reading passages from Readworks.org that averaged 351 words in length with Lexile levels of 770-875 (i.e., fourth-grade range). Lexile levels are measures of readability based on word length and sentence length. The three participants' Lexile levels ranged between 535 and 565 (i.e., third-grade range). We selected passages with fourth-grade Lexile levels in order to promote on–grade level reading comprehension. Baseline reading passages were presented to the participants in their original format from Readworks.org (i.e., in paragraph form with no embedded prompts). For intervention, the experimenter reformatted the reading passages by sectioning them into four parts. After each section, a stopping point prompt was embedded after each sentence with either two lines or a stop sign icon. In each subsequent reading passage during intervention, the two lines were systematically faded (i.e., one at a time) and replaced with a stop sign icon.

The reading/language arts curriculum in the participants' setting included reading comprehension goals for narrative and expository reading passages. Expository passages were selected for this study because students are more likely to struggle with reading comprehension of expository passages compared to narrative passages (e.g., Best, Floyd, & McNamara, 2008; Sainz & Fuchs, 2002).

Generalization probes. To assess generalization, the experimenters used reading passages from the participants' fourth-grade-level science and social studies textbooks. To control for background knowledge, passages were selected that were scheduled to be presented after completion of the study. Similar to baseline passages, the generalization passages were presented to students in their original format with no embedded prompts to self-question.

Comprehension quizzes. The quizzes consisted of 8 experimenter-developed multiple-choice questions pertaining to each reading passage. The first five questions on each quiz addressed the main idea and details, vocabulary, and sequencing. The last three questions addressed the author's purpose, cause and effect relationships, and drawing conclusions. Prior to the implementing the intervention, the language arts teacher examined the quizzes to help determine whether they were approximately equal in difficulty. Quizzes or topics that the teacher identified as too difficult or too easy were removed from the quizzes. There was no set time limit to complete the quiz, but students usually completed the quiz within 5 to 10 minutes.

Definition and measurement of dependent variable

The dependent variable was the number of multiple-choice questions answered correctly (out of 8) on comprehension quizzes. During baseline and generalization, the participants took the quiz immediately after reading the passage. During intervention, the participants took the quiz immediately after the self-questioning intervention session. Responses were scored as correct or incorrect by comparing them to an experimenter-created answer key. Correct answers were those that matched the answers on the answer key. Incorrect answers were those that either did not match the answer key or were left unanswered. The multiple-choice test was untimed, but participants usually took about 5 to 10 minutes to complete it.

Experimental design

A multiple probe across participants design was used to examine the effects of the self-questioning intervention package on the number of multiple-choice comprehension questions answered correctly. The experimental phases were baseline, self-questioning training, and systematic prompt fading. In addition, generalization probes were administered in each phase.

Procedures

Baseline. During baseline, the experimenter worked with all three participants at the same time in a small group. She provided the participants with an expository reading passage and gave the following directions, "Please read the passage silently. When you are finished, you will take a short comprehension quiz." After the participants read the passage silently, the experimenter took the passage from the participants and administered an 8-item multiple choice quiz. After the participants took the quiz, the experimenter thanked them and ended the session. No feedback was provided during baseline.

Self-questioning training. During self-questioning training, the experimenter worked one-on-one with each participant providing direct instruction on generating questions. Instruction included modeling, guided practice, and corrective feedback. First the participant read each paragraph of the passage aloud to the experimenter who provided feedback on miscues if needed. After each paragraph, the experimenter prompted the participant to underline an important sentence (e.g., "Martin Luther King was an important leader in the civil rights movement.") and prompted the participant to change the underlined sentence to a question (e.g., "Who was Martin Luther King?"). The experimenter provided feedback on whether or not the question pertained to the paragraph, whether the answer was correct, and whether the answer could be found in the paragraph.

If a participant struggled with generating a question and answer, the experimenter modeled a correct question and answer, then prompted the participant to say the question and answer. After each training session, the participants took the comprehension quiz. Participant-generated questions were open-ended, while the quiz questions were multiple-choice. After completing the quiz, the experimenter provided praise for correct answers and corrective feedback for incorrect answers. As intended, the participants usually asked questions similar to those presented on the quizzes.

Systematic prompt fading. During this phase, the experimenter prompted the participants to read the passages silently and write their questions and answers on the lines provided after each section of the passage. The experimenter

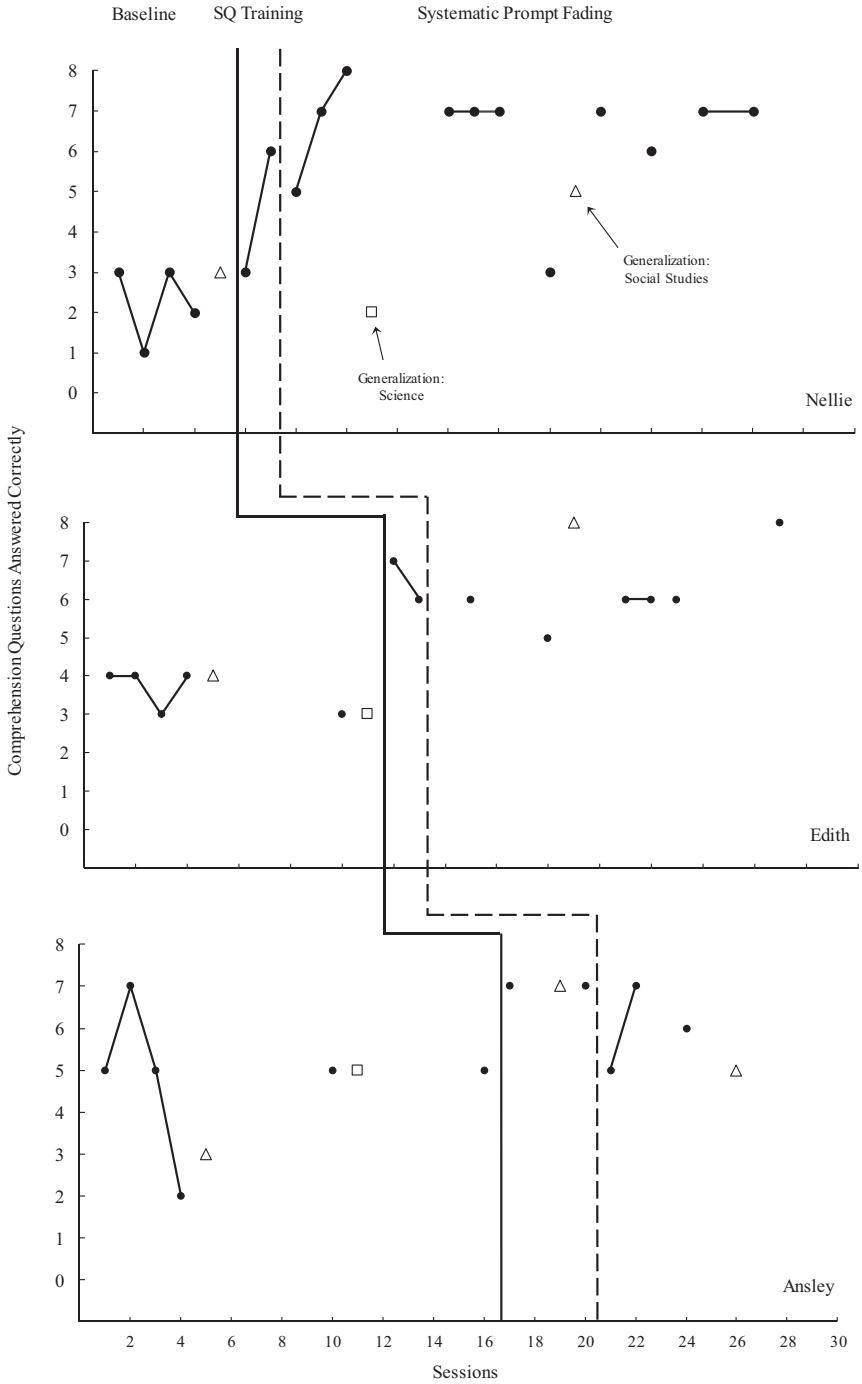


Figure 1. Number of comprehension questions answered correctly by each participant in each condition. Squares represent science generalization probes and triangles represent social studies generalization probes.

gave the participants feedback on the quality of their questions and answers. Then the experimenter administered the comprehension quiz and provided the participants with feedback. On each subsequent prompt fading session on which the participants scored at least 75% correct on the quiz, the experimenter provided the participant with reading passages on which the blank lines (prompts for written responses) were faded one at a time to stop sign icons (prompts for oral responses).

Generalization probes. Generalization passages were selected from the participants' fourth-grade curriculum science and social studies textbooks. The selected generalization passages had not yet been covered in their content area classes during the study. In addition, no prompts to self-question were embedded in the generalization passages. The experimenter administered generalization probes during baseline and intervention for Nellie, Edith, and Ansley. Ansley also completed a generalization probe during training. The experimenter provided the participants with a reading passage and the following verbal directions, "Please read the passage silently. When you are finished, you will take a short comprehension quiz." After the participants read the passage silently, the experimenter took the passage from the participants and administered the quiz.

Table 2. Mean and percentage scores for comprehension quizzes.

Student	Baseline mean	Baseline percent	Train mean	Train percent	Fading mean	Fading percent	Overall percent increase
Nellie	2.25	28%	4.5	56%	6.45	81%	55%
Edith	3.6	45%	6.5	81%	6.2	77%	36%
Ansley	4.8	60%	7.0	88%	6.0	75%	15%

Interobserver agreement (IOA)

IOA was assessed on 57% of the participants' comprehension quizzes across all phases of the study. The experimenter and a second observer independently scored copies of the quizzes using an answer key for each quiz. IOA was examined on a question-by-question basis and calculated by dividing the number of agreements by the number of agreements plus disagreements and multiplying by 100. The total IOA for the participants' post-comprehension quizzes across all phases of the study was 99.8%.

Procedural fidelity

Procedural fidelity data were collected for all participants on 80% of baseline sessions, 51% of training and systematic prompt faded sessions, and 75% of generalization sessions. A data collector observed the experimenter implement the intervention using a checklist for each condition and recorded whether or not the experimenter performed each step correctly and in the correct sequence. Procedural fidelity was calculated by dividing the number of steps completed correctly by the total number of steps and multiplying by 100. Total procedural fidelity was 100% for baseline, 99.9% for training and intervention, and 100% for generalization.

Results

Figure 1 shows the total number of multiple-choice questions answered correctly by each participant in each phase. All three participants demonstrated a similar pattern of responding. Nellie and Edith showed stability in baseline, but Ansley's baseline scores were more variable. During the self-questioning training phase and the systematic prompt fading phase, all three participants immediately increased their level of performance over their baseline levels. Edith's intervention data showed no overlapping data points with baseline, Nellie had one overlapping data point with baseline, and Ansley had five overlapping data points with baseline (due to one high score of 7 correct in baseline).

Table 2 shows the mean number of correct responses on the multiple-choice comprehension quizzes and the mean percentage correct for each participant in each phase. Baseline quiz scores ranged from 1 to 7 multiple-choice comprehension questions answered correctly (mean = 3.7). The self-questioning training phase quiz scores ranged from 3 to 7 correct answers (mean = 6.0). The systematic prompt fading phase quiz scores ranged from 3 to 8 correct answers (mean = 6.3). The generalization quiz scores ranged from 3 to 5 correct answers (mean = 3.6) in baseline and 2 to 8 (mean = 5.4) during systematic prompt fading.

Nellie. Prior to the implementation of the intervention, Nellie had a stable baseline. During baseline, the total number of multiple-choice comprehension questions answered correctly ranged between 1 and 3 with a mean of 2.25 (28%). During the self-questioning training phase, the total number of multiple-choice comprehension questions answered correctly ranged between 3 and 6 with a mean of 4.5 (56%). In the systematic prompt fading phase, the total number of multiple-choice comprehension questions answered correctly ranged from 5 to 8 with one outlier of 3 with a mean of 6.45 (81%). The percentage of non-overlapping data points (PND) between the baseline phase and intervention phases were calculated was 92%.

Edith. Prior to the implementation of the intervention, Edith had a stable baseline. During baseline, the total number of multiple-choice comprehension questions answered correctly ranged between 3 and 4 with a mean of 3.6 (45%). During the self-questioning training phase, correct answers ranged between 6 and 7 with a mean of 6.5 (81%). In the systematic prompt fading phase, multiple-choice comprehension questions answered correctly ranged from 5 to 8 with a mean of 6.2 (77%). There were no overlapping data points between the baseline phase and intervention phases for Edith; PND was 100%.

Ansley. Prior to the implementation of the intervention, Ansley's baseline data were variable. During baseline, multiple-choice comprehension questions answered correctly ranged between 2 and 7 with a mean of 4.8 (60%). During the self-questioning training phase, questions answered correctly increased to 7 with a mean of 7 (88%). During systematic prompt fading, questions answered correctly ranged from 5 to 7 with a mean of 6 (75%). PND between the baseline phase and intervention phases was 0%.

Discussion

The purpose of this study was to examine the effects of a self-questioning strategy with systematic prompt fading on the reading comprehension of three fourth-graders identified as at-risk. The systematic prompt fading procedure was used to assist participants with moving from writing self-questions (blank lines) to generating self-questions orally (stop signs). A multiple probe across participants design demonstrated improvements in reading comprehension for all three participants. This supports previous research on the effectiveness of self-questioning for improving reading comprehension of students with learning disabilities (e.g., Crabtree et al., 2010; Rouse et al., 2014) and extends the findings to at-risk fourth graders. Nellie showed the greatest improvement, with a 55% mean increase over baseline, and Ansley showed least improvement, with a 15% mean increase over baseline.

During the generalization probes, all participants demonstrated at least some generalization of self-questioning and reading comprehension to the higher-level reading passages in their science and social studies textbooks (i.e., fifth- to seventh-grade reading level). Edith demonstrated the most generalization with an increase of 56% from baseline to intervention. In addition, she achieved 100% correct on one of the generalization probes. Ansley demonstrated minimal generalization with an increase of 25% from baseline to intervention. Nellie demonstrated the least generalization, with an increase of 6% from baseline scores to intervention. Considering that the participants were fourth graders reading below grade level, their demonstration of generalized outcomes with advanced reading materials is significant.

The positive results of this intervention can be attributed to learning a strategy that required their active responding to the text. Training participants to ask and answer questions likely helped them identify relevant information. Systematically fading the questions from a written to an oral format may have facilitated the transition to increased independence of self-monitoring reading comprehension.

Limitations and future research

Although this study supports previous systematic prompt fading and self-questioning research, the experimenters identified several limitations. One limitation is that comprehension was measured using a multiple-choice format, which may have inflated participant scores if they correctly guessed an answer they did not know. Future research should attempt to include comprehension assessments that measure recall (in addition to or instead of recognition) by using open-ended questions, retelling a passage, or writing a summary.

Another limitation related to assessment was an absence of a formal measure for the types of questions the participants generated and the quality of their questions and answers. Future research should attempt to assess the quality and type of self-generated questions and answers to determine whether these factors influence reading comprehension. Student-generated questions may be assessed using a rubric that analyzes how well the questions and answers captured important points; whether the questions were literal, inferential, or critical; and how similar the student-generated questions are to the comprehension quiz questions.

An important limitation that is probably common to most reading comprehension research is that experimenters were unable to control for the participants' background knowledge of topics in the selected reading passages. It is possible that background knowledge or exposure to other topics during the study contributed to the variability of the data. For example, Ansley scored 7 out of 8 correct on her second baseline session, which may be an indication that she had background knowledge of the reading passage topic used on that session. Future research should attempt to control for background knowledge or exposure to other topics by consulting with teachers and/or examining curriculum materials. Future research could also pre-assess the participants' knowledge of specific topics prior to selection.

Other limitations were related to the generalization. Although this study was designed to experimentally measure generalization, more generalization probes would provide further evidence of the extent to which the participants achieved generalization to more advanced reading passages in their textbooks. In addition, generalization was limited to reading passages in expository textbooks. Future research could extend generalization measures to other settings (e.g., content area classrooms, library), other types of reading materials (e.g., narrative texts, persuasive texts, biographies), and other types of text formats (e.g., computer- or web-based, newspapers, magazines).

Due to time restrictions (i.e., end of school year), the experimenters were unable to assess maintenance of the self-questioning strategy over time. Future research would be enhanced by implementing a maintenance phase. In addition, due to time constraints, social validity was not assessed. Future research should also include a social validity measure to determine the acceptability of the intervention by participants, teachers, and parents.

Implications for practice

The self-questioning strategy is easy to implement, time-efficient, cost-effective and, most important, increases student independence. Students who are easily distracted or unaware of how much they are comprehending can benefit from learning to self-monitor their reading comprehension by asking themselves questions periodically while they are reading. Self-questioning can be useful for a wide range of individual learning needs as it allows for flexibility of reading levels, genres, and prompt fading procedures. This flexibility can be very useful for teachers who, for example, use multi-tiered models of instruction in their classrooms.

For students who need more support and structure, teachers may start the intervention by providing generic questions and gradually transitioning to self-generated questions. In addition, teachers can prompt students to ask and answer questions more frequently throughout the passage. Rather than having four stopping points, for example, students could have six or eight stopping points. Alternatively, students with higher cognitive abilities may benefit from learning how to ask inferential and critical thinking questions. Teachers can also implement self-questioning interventions using peer-mediated instruction in which students work with partners asking each other questions about what they are reading together. Students may find this instructional arrangement especially motivating.

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